Summary and Recommendation

This project analyzes the Telco Customer Churn dataset to understand customer behavior and identify the major factors influencing churn. The dataset contains **7,043 customer records** and includes demographic information, service subscriptions, account details, and churn status.

During data preparation, the `TotalCharges` field was converted into numeric format to make it suitable for analysis. The dataset showed **no significant missing values and no duplicate customer IDs**, which indicates that the data quality is reliable enough for meaningful business insights.

Overall churn performance

The analysis shows that **29.54% of customers have churned**, while **70.46% have been retained**. In practical terms, this means that **nearly 3 out of every 10 customers leave the company**. From a business perspective, this is a significant level of customer loss because acquiring new customers generally costs more than retaining existing ones.

Gender analysis

The gender-based charts indicate that churn is **fairly evenly distributed between male and female customers**. The proportion of male and female customers in the dataset is nearly balanced, and the churn pattern across both groups does not show a major difference.

Key insight: gender does not appear to be a strong predictor of churn.

Senior citizen analysis

The senior citizen segment shows a **higher tendency to churn compared with non-senior customers**. Although senior citizens represent a smaller portion of the customer base, their churn proportion is comparatively higher.

This suggests that older customers may face challenges related to pricing, service usability, or support accessibility.

Business implication: targeted support and simpler service communication for senior customers could improve retention.

Tenure analysis

Tenure emerged as one of the strongest indicators of churn.

The charts show that customers with **low tenure (especially new customers in the first few months)** are much more likely to leave. In contrast, customers who remain with the company for a longer period become increasingly stable and are less likely to churn.

This means that the **highest churn risk occurs during the early stage of the customer lifecycle**.

Interpretation: if customers are not satisfied early, they are more likely to discontinue the service before building long-term loyalty.

Contract type analysis

Contract type has a very strong relationship with churn.

- **Month-to-month customers have the highest churn**
- **One-year contract customers churn much less**
- **Two-year contract customers show the lowest churn**

The charts clearly indicate that customers with longer contractual commitment tend to stay longer.

Business insight: customers with flexible short-term contracts can leave more easily, while longer contracts improve retention and predictability.

Internet service analysis

The internet service charts reveal a clear pattern:

- **Fiber optic customers show noticeably higher churn**
- **DSL customers show comparatively lower churn**
- Customers without internet service show lower churn levels

A likely explanation is that customers using fiber optic services may have **higher expectations regarding performance, pricing, or service quality**, which can influence satisfaction.

Value-added service analysis

The analysis of additional services shows that churn is higher among customers who **do not use support-related or protection-related services**, including:

- Online security
- Online backup
- Device protection
- Tech support

Customers subscribed to these services tend to remain more stable.

Interpretation: additional services increase perceived value and strengthen customer attachment to the company.

Payment method analysis

Payment method also shows an important behavioral pattern.

Customers using **electronic check** have the **highest churn rate** compared with customers using:

- Mailed check
- Bank transfer
- Credit card

This suggests that payment behavior may be linked to customer commitment, convenience, or satisfaction.

Monthly charges and total charges

The charts suggest that customers with **higher monthly charges tend to churn more frequently**. Customers paying higher recurring fees may become more sensitive to pricing and may compare competing alternatives more actively.

At the same time, customers with **higher total charges often show lower churn**, largely because total charges generally increase with longer tenure. In other words, long-term customers accumulate higher total spending and also tend to be more loyal.

Key findings in numbers

- **Total customers:** 7,043
- **Retained customers: 70.46%**
- **Churned customers: 29.54%**
- **Approximate interpretation:** nearly **1 out of every 3 customers leaves**

Business conclusion

The analysis indicates that churn is concentrated among customers who are:

- **New customers with low tenure**
- **Senior citizens**
- **Month-to-month subscribers**
- **Fiber optic users**
- **Customers not using online security, backup, device protection, or tech support**

- **Electronic check users**
- **Customers paying relatively higher monthly charges**

These factors together suggest that churn is driven mainly by **early customer experience, contract flexibility, pricing sensitivity, and lower service engagement**.

Strategic recommendations

Based on the analysis, the company should focus on:

1. Improve early-stage customer onboarding

Since churn is highest among new customers, the first few months are critical. Better onboarding, guidance, and proactive support can improve early retention.

2. Encourage long-term contracts

Providing incentives for one-year and two-year plans may reduce churn among month-to-month users.

3. Promote value-added services

Bundling online security, tech support, backup, and device protection can improve customer stickiness.

4. Review fiber optic customer experience

Investigate pricing, service quality, and customer satisfaction for fiber optic users.

5. Target high-risk customer groups

Special retention campaigns can be designed for senior citizens, new customers, and customers using electronic check.